# Week 3 — Multi-Agent Systems and Emergent Behavior

## Theme

Week 1 taught us that simple rules can generate complexity.

Week 2 taught us that learned local rules can grow and repair structures.

Week 3 explores a deeper question:

**What happens when many independent entities interact with each other?**

Ant colonies build nests.

Birds form flocks.

Fish form schools.

Traffic jams emerge.

Crowds self-organize.

Nobody is in charge.

Yet coordinated behavior appears.

This phenomenon is called **emergence**.

By the end of this week, you should understand why decentralized systems can exhibit intelligent behavior without any central controller.

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# Learning Goals

By the end of Week 3, you should be able to explain:

1. What is an agent?
2. What is a multi-agent system?
3. What is local information?

4. Why do global patterns emerge from local interactions?
5. How does information spread through populations?
6. Why can societies form without a leader?

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# Required Resources

## Resource 1 — Boids

Watch:

Sebastian Lague — Boids

https://www.youtube.com/watch?v=bqtqltqcQhw

Goal:

Understand how three simple rules create flocking behavior.

While watching, answer:

- Why do birds appear coordinated?
- Is there a leader?
- What information does each bird have?

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## Resource 2 — Original Boids Paper

Read:

Craig Reynolds — Flocks, Herds, and Schools

https://www.red3d.com/cwr/papers/1987/boids.html

Focus on:

- Separation
- Alignment
- Cohesion

Do not worry about mathematics.

Try to understand the intuition.

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# Resource 3 — Emergence

Watch:

Kurzgesagt — Emergence

https://www.youtube.com/watch?v=16W7c0mb-rE

Goal:

Understand how complexity emerges from simple interactions.

Think about:

- Ant colonies
- Human societies
- Markets
- Neural networks

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# Resource 4 — Agent-Based Modeling

Read:

Introduction to Agent-Based Modeling

https://www.comses.net/resources/tutorials/

Goal:

Understand:

- Agent
- Environment
- State
- Interaction
- Update

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## Resource 5 — Information Propagation

Watch:

Network Effects and Spreading Dynamics

https://www.youtube.com/watch?v=JPnqYFyj9J4

Goal:

Understand how information spreads through a network.

Think about:

- Rumors
- Disease
- Trust
- Panic

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# Practical Assignment

## Task 1 — Implement Boids

Build a Boids simulation.

Rules:

1. Separation
2. Alignment
3. Cohesion

Requirements:

- Minimum 100 agents
- Real-time visualization
- Adjustable parameters

Submission:

- GitHub repository
- 30-second video

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## Task 2 — Parameter Exploration

Run experiments.

Change:

- Neighbor radius
- Alignment strength
- Separation strength

Observe:

- Stable flocking
- Fragmentation
- Oscillation

Submission:

A one-page report containing:

- Parameter values
- Screenshots
- Observations

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## Task 3 — Obstacles

Add obstacles to the environment.

Examples:

- Walls
- Circular barriers
- Safe zones

Observe:

How does the flock adapt?

Submission:

30-second video.

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## Task 4 — Predator and Prey

Introduce:

- Predator agents
- Prey agents

Rules:

- Predators chase
- Prey avoid

Observe:

- Group formation
- Escape behavior
- Splitting and merging

Submission:

Video and written observations.

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## Task 5 — Information Spread

Select one agent.

Mark it as "informed."

Goal:

Visualize how information spreads through the population.

Examples:

- Danger signal
- Food discovery
- Evacuation signal

Submission:

Video and explanation.

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# Reflection Questions

Answer the following:

1. Why does flocking emerge?
2. Why is no leader required?
3. What happens when perception radius becomes very small?
4. What happens when perception radius becomes very large?
5. What causes a flock to fragment?
6. How is Boids similar to Cellular Automata?
7. How is Boids different from Neural Cellular Automata?
8. What examples of emergent behavior exist in the real world?

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# Mini Project

Zombie Evacuation Simulator

Agents:

- Civilians
- Zombies

Rules:

- Civilians move away from zombies.
- Zombies move toward civilians.
- Civilians can share danger information.

Metrics:

- Survival rate
- Number of groups formed
- Average group size
- Time until collapse

No machine learning.

No neural networks.

Only local rules.

The goal is to observe emergent behavior.

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# Submission Checklist

Required:

- Boids implementation
- Obstacle simulation
- Predator-prey simulation
- Information propagation simulation
- Reflection answers
- Zombie evacuation simulator

Optional:

- Interactive controls
- Parameter sliders
- Performance optimizations

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# Pass Criterion

A student passes Week 3 if they can explain:

"Complex global behavior can emerge from simple local interactions without centralized control."

This idea is the foundation of:

- Cellular Automata
- Neural Cellular Automata
- Multi-Agent Systems
- Swarm Intelligence
- Trust Networks
- Societies
- The final Containment Protocol project